

UNIVERSITY OF SCIENCE
VIETNAM NATIONAL UNIVERSITY, HCM CITY
INFORMATION TECHNOLOGY FALCULTY

INTRODUCTION TO SOFTWARE ENGINEERING

PA0 - Project idea & Team contract

Team ID: T09

Presented by

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24127565 - Nguyen Hoang My Trang

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24127322 - Phan Nhat Anh

Lecturers

Nguyen Minh Huy

Nguyen Le Hoang Dung

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1 Team Information

Item	Value
Team ID	T09
Team Name	TetraBo
Team Lead	Nguyen Hoang My Trang
Communication Channel	Discord

Student ID	Full Name	Email	Main Role	Backup Role
24127565	Nguyen Hoang My Trang	nhmtrang2415@clc.fitus.edu.vn	Team Lead / PM	Developer
24127497	Doan Hong Phuc	dhphuc2429@fitus.clc.edu.vn	BA / Designer	Team Lead / PM
24127079	Nguyen Tien Minh	ntminh2435@clc.fitus.edu.vn	Developer	Database / CI
24127322	Phan Nhat Anh	pnhatanh2437@clc.fitus.edu.vn	Tester	Developer

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

CampusBorrow – a web application that lets students and club representatives submit equipment loan and room booking requests, while giving resource managers a single interface to approve, hand over, and record returns.

2.2 Problem and Motivation

Campus units currently manage equipment loans and room bookings through Google Forms, chat messages, and spreadsheets. This leads to:

- No real-time view of equipment availability or room schedules.
- Two students can request the same item simultaneously with no conflict check.
- Equipment status must be updated manually - prone to stale data and errors.
- Overdue loans surface only when a manager audits the spreadsheet.
- Approval history (who approved, when, why rejected) is not structured.
- Room double-bookings occur when requests come through separate channels.

CampusBorrow consolidates catalogue, request lifecycle, and history into one role-based system.

2.3 Target Users

User Group	What They Need from the System
Borrower (Student / Club Representative)	Browse equipment and room availability; submit, track, and cancel requests; view personal history.
Resource Manager	Process requests for assigned equipment/rooms; record hand-over and return; monitor overdue loans.
Admin	Create/deactivate accounts; assign roles and resources; view all requests and history unit-wide.

2.4 Main Features

Must-have features (F01–F09) must be fully implemented by PA5. F10 is a nice-to-have attempt only if scope is completed early.

#	Feature	Short Description	Must-have?
F01	Authentication & RBAC	Three roles: Borrower, Resource Manager, Admin. Accounts created by	Yes

		Admin only; all permission checks on the server.	
F02	Equipment & Room Catalogue	Searchable list of equipment and rooms. Managers edit assigned resources; Admin manages all. Rooms show capacity and availability.	Yes
F03	Equipment Borrow Request & Approval	One item per request; requester states purpose and return date. Manager approves or rejects; approval reserves the item atomically.	Yes
F04	Hand-over, Return & Overdue	Manager records hand-over and return times; system updates availability. Loans past deadline appear in an overdue list.	Yes
F05	Room Schedule View	Shows approved booking slots for a chosen room on a given day so requesters can find a free slot.	Yes
F06	Room Booking Request	One room, one day; requester enters start/end time, purpose, and headcount. The system checks time order, future date, and capacity.	Yes
F07	Room Booking Approval & Conflict Check	Manager approves or rejects; approval blocks if an approved slot overlaps. Requesters can cancel before start time.	Yes
F08	Request & History Tracking	Borrowers see their own requests; Managers see their scope; Admin sees all. Every action logged with actor, outcome, reason, and time.	Yes
F09	Global Administration	Admin creates/deactivates accounts, assigns roles, assigns resources to managers. Deactivation does not erase history.	Yes
F10	Usage Statistics & CSV Export	Usage summary per resource; export borrow/booking list to CSV.	No – Nice-to-have

2.5 Similar Products

Snipe-IT (snipe-it.readme.io): Full IT-asset management with multi-location, depreciation, and LDAP. CampusBorrow is scoped to one unit, two resource types, and a simple three-role approval flow.

MRBS – Meeting Room Booking System (mrbs.sourceforge.io): Room-only scheduling, often self-service. CampusBorrow combines equipment and room booking under a manager-approval workflow suited to shared academic resources.

Google Forms + Spreadsheet: The current practice in most campus clubs. No real-time availability, no conflict prevention, no structured audit trail. CampusBorrow directly replaces this workflow.

2.6 Platform and Technology Ideas

Not final; to be confirmed at PA1.

- Platform: web application (responsive); no separate mobile app.
- Frontend: React.
- Backend: Node.js / Express (or equivalent).
- Database: PostgreSQL – transactions and conditional unique constraints for conflict prevention.
- Version control: Git / GitHub with pull requests and CI (GitHub Actions).
- Deployment: Render or Railway for the demo; finalised at PA1.

2.7 Feasibility, Scope and Open Questions

Feasibility: F01–F09 are primarily CRUD flows plus two state machines (equipment lifecycle, room booking lifecycle). The hardest part is the atomic approval-and-reserve step (F03/F07). With 4 members working 8–10 h/week over three coding weeks (Wk 8–10), the scope is feasible if the team finishes the core borrow/booking flow before auxiliary features.

Out of scope:

- academic timetable integration; event registration;
- recurring bookings;
- multiple items/rooms per request;
- advance equipment scheduling; waitlists;
- maintenance tracking; monetary transactions;
- automated email/SMS; SSO; multi-campus deployment;
- native mobile app.

Main risks:

- (1) two managers approving the same resource simultaneously → use DB transactions;
- (2) time-boundary edge cases → standardise on UTC+7, test all boundaries;
- (3) scope creep → freeze Must-have list after PA1.

Questions for the TA:

- To what extent is the team using AI tools (Copilot, Claude, Cursor...) to generate code? Are there any limits? How detailed do we need to record it in the AI Usage Log?

3 Team Contract

3.1 Goals and Expectations

- Proposed goal: get a project score of 10/10, complete all Must-have items, have a working deployed version, and be able to explain the group's decisions. This is a goal, not a guarantee of the instructor's grade.
- Each person is expected to spend 8–10 hours/week during the documentation phase, and 12 hours/week during the implementation and testing phase. The group will reconfirm this capacity at PA1.
- Members should inform early about weeks with exams, personal matters, or reduced time; the group will adjust tasks before it affects the deadline.
- Everyone must understand their own work and be able to explain other parts of the group's work as required by the course.
- Contributions are recorded based on actual work and evidence; it is not assumed that everyone automatically gets 25%.

3.2 Communication

- Main channel for discussion: Discord.
- Expected response time: within 24 hours on weekdays.
- Decisions and files kept in: GitHub repository and GitHub Project task board.

3.3 Meetings

- Weekly TA status meeting: day and time to be confirmed with the TA at PA1 (online, ~15 min, every **9:00 am, Tuesday**).
- Internal team meeting: every **09:00 am, Wednesday** – review progress, blockers, decisions.
- Attendance rule: notify the team at least 12 hours in advance if unable to attend; absent members send a written update and read minutes within 24 hours. Missed meetings are recorded in the weekly status report.
- Weekly status report deadline: every **Sunday** at **22:00 pm** – committed to repository.

3.4 Roles and Responsibilities

All members code, review, and test. Roles define primary accountability for non-code deliverables. Each backup reviews PRs for the primary so no one reviews their own work.

Role	Primary	Backup	Extra Responsibilities
Team Lead / PM	Nguyen Hoang My Trang	Doan Hong Phuc	Project plan, task board, meeting facilitation, weekly reports, PA coordination.
BA / Designer	Doan Hong Phuc	Nguyen Hoang My Trang	Stakeholder interviews, use cases, requirements, UI mockups, ERD.
Developer	Nguyen Tien Minh	Phan Nhat Anh	Project scaffolding, GitHub/CI setup, auth & access control backend, deployment.
Tester	Phan Nhat Anh	Nguyen Tien Minh	Test plan, test cases, automated test suite, defect log.

3.5 Work Standards

- Internal deadline: each member finishes their part 12 hours before the PA submission deadline.
- Quality: every document section and pull request is reviewed by at least one other member before submission.
- Tools: every task is on the task board with an owner and due date; code goes through pull requests.

3.6 Use of AI Tools

- Permitted tools: ChatGPT, GitHub Copilot, Claude, or equivalent – for drafting, explanation, code suggestion.

- Each member is responsible for reviewing and verifying any AI output before use in a deliverable.
- AI usage log: each member records their own usage.
- The course GenAI policy applies. Undisclosed AI use is an academic integrity violation.

3.7 Decision Making

- Default: consensus after all members discuss options and evidence.
- If no consensus: simple majority vote (3 of 4). In case of a tie, the PM decides and records the reason.
- Decisions on Must-have scope, PA deadlines, or baselines require all available members to review first.

3.8 Handling Conflicts and Missed Work

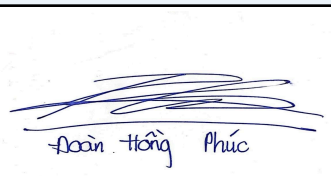
Step 1: Talk directly – clarify cause and impact, agree on a fix and a new date; PM monitors.

Step 2: Discuss in a team meeting – redistribute work if needed; record in the weekly status report.

Step 3: Inform the TA – PM provides evidence (tasks, PRs, commits); individual contribution factor adjustment is at the TA's / lecturer's discretion.

3.9 Signatures

By signing, each member agrees to follow this contract.

Student ID	Full Name	Signature	Date
24127565	Nguyen Hoang My Trang		October 14th, 2026
24127497	Doan Hong Phuc		
24127079	Nguyen Tien Minh		

24127322	Phan Nhat Anh		

4 AI Usage Log

[Required in every PA. List each significant use of AI tools. Write "No AI tools were used" if none. Undisclosed AI use is an academic integrity violation.]

#	Member	Tool	Task / Purpose	How Output Was Verified and Changed
1	Nguyen Hoang My Trang	Claude (Anthropic)	Drafting PA0 – Project Idea and Team Contract sections.	The draft generated by AI has been read by the team and personal information has been added to the report, while also checking if the content matches the information the team had agreed on before.